Muhammad Rehan Ul Haq

Senior Software Engineer | Fintech & Payment Systems

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PROFESSIONAL SUMMARY

Senior Software Engineer with 4+ years of experience building secure, reliable fintech and banking systems in Java and Spring Boot. Builds payment infrastructure end to end: a RAAST payment switch integrated with the State Bank of Pakistan (P2P and P2M), ISO 20022 message processing, and a configuration-driven ISO 20022 simulator for integration testing, alongside digital banking platforms, digital wallets, and enterprise CRM for tier-1 banks. Strong in service architecture design, design patterns, and service optimization. Leads teams, reviews code, and mentors developers in Agile environments. Open to remote backend and fintech engineering roles.

PROFESSIONAL EXPERIENCE

Senior Software Engineer | **Vaulsys** *Feb 2026 – Present*

Karachi, Pakistan

RAW Bank – Agent Portal & Middleware

- Lead the Agent Portal team for RAW Bank's digital banking platform, owning backend development of the portal and its middleware services in Java and Spring Boot.
- Build secure RESTful APIs and microservices for high-volume financial transactions, covering authentication, authorization, and integration of core banking workflows.
- Drive system design and architecture decisions and run code reviews to keep banking services performant, reliable, and maintainable.

RAAST Payment Switch – external client

- Developed a RAAST payment switch that connects a client institution to the State Bank of Pakistan's RAAST instant payment system, implementing Person-to-Person (P2P) and Person-to-Merchant (P2M) payment flows.
- Implemented ISO 20022 XML message processing (including pacs.008 credit transfers, pain.013 request-to-pay for RTP Now and RTP Later, and admi.002 rejections) with request/response handling, transaction lifecycle tracking, and error handling.
- Handled XML digital signatures (XAdES, RSA-SHA256) and built Oracle-backed request and transaction logging through stored procedures.
- Diagnosed and resolved SBP schema-validation rejections during system integration testing, correcting message construction for mandatory and conditional fields.

Configurable ISO 20022 Simulator

- Designed and built a configuration-driven ISO 20022 simulator that replaces hardcoded message structures with configurable message templates, field mappings, and incoming/outgoing message flows.
- Implemented field-level validation with configurable regular expressions, data types, and length constraints, plus echo-back mapping of fields between related request and response messages.
- Added optional Business Application Header (AppHdr) and signature (Sgntr) handling and reusable scenario-based simulation workflows that make payment integrations easier to test and validate.

Senior Software Engineer | **Avanza Solutions** *Nov 2025 – Jan 2026*

Karachi, Pakistan

- Developed, enhanced, and supported enterprise-grade CRM solutions for Meezan Bank, one of the organization's largest clients.
- Designed and implemented scalable backend services in Java and Spring Boot with a focus on performance and maintainability.
- Led development of new features, third-party system integrations, and banking service integrations; resolved production issues and optimized existing functionality.
- Translated business requirements into technical solutions with cross-functional teams and client stakeholders.
- Contributed to system architecture discussions, conducted code reviews, and mentored junior developers on industry best practices.

Senior Software Engineer | **Paysys Labs Pvt Ltd** *Jul 2025 – Oct 2025*

Karachi, Pakistan

- Worked on RAAST P2M (Person-to-Merchant), Pakistan's real-time payment infrastructure, enabling seamless merchant transactions and improving payment efficiency.
- Integrated P2M services and played a key role in designing P2P and instant transfer services using ISO financial messaging standards.
- Implemented and optimized push notification systems for timely customer communication on critical transactions and alerts.
- Developed digital banking solutions that improved customer access to banking services.

Software Engineer | **Paysys Labs Pvt Ltd** *Jun 2023 – Jun 2025*
Karachi, Pakistan

- Developed core modules of the PAYVAY Digital Wallet (PSO): customer onboarding, NADRA biometric verification, payment processing, and bill payments.
- Enhanced existing middleware services and built new application services aligned with evolving project requirements.
- Engineered RESTful APIs with Java, Spring Boot, and JPA; deployed microservices with Git, Maven, and Docker in Agile sprints with a focus on performance and scalability.

Software Engineer | **Irenic Tech** | **Freelance Java Developer** *Feb 2022 – May 2023*
Karachi, Pakistan

- Engineered scalable RESTful APIs with Java, Spring Boot, and JPA for high-traffic fintech applications.
- Implemented performance optimizations and applied SOLID principles and design patterns to improve maintainability.
- Integrated OneSignal APIs for targeted push notifications across mobile and back-office platforms.
- Built and deployed microservices in an Agile team using Git, Maven, and Docker for CI/CD.

TECHNICAL SKILLS

Programming Languages: Java, SQL

Backend Development: Spring Boot, Spring Security, Spring Cloud Gateway, JPA, Hibernate, REST APIs, Microservices

Architecture & Design Patterns: Service Architecture Design, Command Registry Pattern, Microservices Architecture, Distributed Systems, API Design, Design Patterns, SOLID Principles, Scalability, High Availability, Load Balancing

Performance & Code Quality: Service Optimization, Performance Optimization, Code Reviews, Refactoring, JUnit, Mockito

Banking & Payment Systems: RAAST (P2P, P2M, Request-to-Pay), Payment Switch Development, Core Banking Integration, Digital Wallets, NADRA Biometric Verification

Messaging Standards & Integration: ISO 20022, ISO 8583, RabbitMQ, REST, Swagger, Postman

Databases: Oracle (incl. stored procedures), MySQL, PostgreSQL, Redis

Security & Auth: JWT, OAuth2, Spring Security, XML Digital Signatures (XAdES), HTTPS, CORS

DevOps & Tools: Docker, CI/CD, Jenkins, GitHub Actions, Linux, Nginx, Maven, Git, GitLab, GitHub, Bitbucket, JIRA, Agile/Scrum

CERTIFICATIONS

Java (Basics) — HackerRank *Aug 2024*
Core Java fundamentals: OOP, collections, control structures, and exception handling demonstrated through timed coding challenges.

Dockerization & CI/CD *Jan 2024*
Docker containerization (Dockerfiles, Compose) and CI/CD pipelines using Jenkins and GitHub Actions.

Spring Boot 3 & Hibernate *Aug 2019*
Backend application development with Spring Boot and Hibernate: REST APIs, JPA, and scalable development practices.

KEY ACHIEVEMENTS

- Built an SBP-connected RAAST payment switch (P2P and P2M) and a configurable ISO 20022 simulator at Vaulsys.
- Led the Vaulsys Agent Portal team delivering high-volume transaction capabilities for RAW Bank digital banking.
- Implemented RAAST P2M real-time payment infrastructure at Paysys Labs, enabling seamless person-to-merchant transactions for thousands of merchants across Pakistan.
- Developed the PAYVAY Digital Wallet (PSO), integrating NADRA biometric verification, payment processing, and bill payments for a nationwide user base.
- Built enterprise CRM at Avanza Solutions for Meezan Bank, Pakistan's largest Islamic bank, delivering new features, third-party integrations, and production support.

EDUCATION

MS — Software Engineering *Aug 2024 – May 2026*
National University of Computer and Emerging Sciences (FAST-NUCES) | *Karachi, Pakistan*
Coursework: *Software Requirements Engineering, Software Architecture, Product Management, System Design, Cloud-Native Architectures, Distributed Systems*

BS — Computer Science *Jan 2019 – Feb 2023*
DHA Suffa University | *Karachi, Pakistan*
Coursework: *Software Requirements Engineering, Software Architecture, Data Structures & Algorithms, Database Systems*